

Here is the complete, finalized engineering blueprint for your 50-meter, inter-building rooftop irrigation system. This consolidates the 1S parallel battery architecture, the 433MHz LoRa point-to-point link, and the spring antenna layout rules into one master document. I have included estimated local market prices (in BDT) so you can easily budget for the parts on Daraz or RoboticsBD.

Phase 1: The Master Bill of Materials (BoM) & Budget

Component	Engineering Purpose	Est. Price (BDT)
ESP32-C3 SuperMini (x2)	1 for Roof Transmitter, 1 for Indoor Receiver	600
RA-02 SX1278 433MHz (x2)	LoRa radio modules (includes spring antennas)	800
12V 1.5W Solar Panel	Polycrystalline outdoor energy harvester	350
LM2596 Buck Converter	Drops fluctuating 12V solar down to a stable 5V	120
TP4056 Charging Module	Safely manages CC/CV lithium battery charging	50
18650 Li-ion Battery (x4)	2500mAh Grade A cells for 10,000mAh reserve	1,200
MT3608 Boost Converter	Steps the 3.7V battery up to 12V for the solenoid	100
12V DC Solenoid Valve	1/2-inch normally closed water shutoff	500
5V 1-Channel Relay	Opto-isolated switch for the 12V solenoid line	100
Capacitive Soil Sensor v1.2	Measures soil moisture without rusting	150
1N4007 Diode	Prevents inductive kickback from destroying circuit	5
IP67 ABS Junction Box	UV-stabilized waterproof housing for the roof node	300
PG7/PG9 Cable Glands (x3)	Creates watertight seals where wires enter the box	100
Total Estimated Budget		~ 4,375 BDT

Phase 2: Complete Connection & Wiring Diagram

1. The Power & Charging Rail

1. **Solar Panel (+) and (-)** \rightarrow Connect to **LM2596** IN+ and IN-.
2. **LM2596 Tuning:** Turn the screw until the output is exactly **5.0V**.
3. **LM2596 OUT+ and OUT-** \rightarrow Connect to **TP4056** IN+ and IN-.
4. **18650 Batteries (x4):** Solder all four in parallel (all Positives together, all Negatives together). *Ensure all cells are at exactly the same voltage before joining!*
5. **Battery Pack** \rightarrow Connect to **TP4056** B+ and B-.

2. The Microcontroller & Radio Logic (Both Roof & Indoor Nodes)

Wire the RA-02 to the ESP32-C3 SuperMini using this strict SPI pinout:

- **Battery B+** \rightarrow ESP32-C3 **5V** pin (The internal LDO will safely drop this to 3.3V).
- **Battery B-** \rightarrow ESP32-C3 **GND**.
- RA-02 3.3V \rightarrow ESP32-C3 3V3 (*Never use 5V for LoRa*)
- RA-02 GND \rightarrow ESP32-C3 GND
- RA-02 NSS \rightarrow ESP32-C3 GPIO 7
- RA-02 MOSI \rightarrow ESP32-C3 GPIO 6
- RA-02 MISO \rightarrow ESP32-C3 GPIO 5
- RA-02 SCK \rightarrow ESP32-C3 GPIO 4
- RA-02 RST \rightarrow ESP32-C3 GPIO 2
- RA-02 DIO0 \rightarrow ESP32-C3 GPIO 3

3. Sensors & Solenoid Actuation (Roof Node Only)

1. **Moisture Sensor:** VCC to ESP32 **3V3**, GND to **GND**, AOUT to ESP32 **GPIO 0**.
2. **MT3608 Boost Converter:** Connect VIN+ and VIN- directly to the Battery B+ and B-. Tune the output screw to exactly **12.0V**.
3. **Relay Logic:** Connect Relay VCC to ESP32 **5V**, GND to **GND**, and IN to ESP32 **GPIO 10**.
4. **Solenoid Switching:** Connect MT3608 VOUT+ (12V) to Relay **COM**. Connect Relay **N.O.** to Solenoid (+). Connect Solenoid (-) to MT3608 VOUT- (GND).
5. **Protection:** Solder the **1N4007 Diode** across the Solenoid (+ and -) terminals, with the silver stripe pointing toward the Relay.

Phase 3: The C++ Production Code

Node A: Rooftop Transmitter (Battery/Solar)

Upload this to the rooftop ESP32. It uses SF10 for excellent 50m penetration and enters deep sleep to preserve the parallel battery bank.

```
#include <SPI.h>
#include <LoRa.h>
#include <esp_sleep.h>
```

```
#define SS_PIN    7
#define RST_PIN   2
#define DIO0_PIN  3
#define SENSOR_PIN 0
#define RELAY_PIN 10
```

```
void setup() {
    Serial.begin(115200);
    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW); // Failsafe: Keep valve closed
```

```

    LoRa.setPins(SS_PIN, RST_PIN, DIO0_PIN);
    if (!LoRa.begin(433E6)) {
        while (1);
    }

    // LoRa Tuning for 50m Urban Penetration
    LoRa.setSpreadingFactor(10);
    LoRa.setSignalBandwidth(125E3);
    LoRa.setTxPower(17);

    // Read and Map Soil Moisture
    int rawMoisture = analogRead(SENSOR_PIN);
    // Note: Adjust 3500 (Dry) and 1500 (Wet) to your specific soil
tests
    int moisturePercent = map(rawMoisture, 3500, 1500, 0, 100);
    moisturePercent = constrain(moisturePercent, 0, 100);

    // Broadcast Packet
    String payload = "ROOF_NODE:" + String(moisturePercent);
    LoRa.beginPacket();
    LoRa.print(payload);
    LoRa.endPacket();

    // 3-Second Window to listen for Home Assistant commands
    unsigned long windowStart = millis();
    while (millis() - windowStart < 3000) {
        int packetSize = LoRa.parsePacket();
        if (packetSize) {
            String incomingCommand = "";
            while (LoRa.available()) { incomingCommand +=
(char)LoRa.read(); }

            if (incomingCommand == "START_WATERING") {
                digitalWrite(RELAY_PIN, HIGH);
                delay(1800000); // 30 minutes (1,800,000 ms)
                digitalWrite(RELAY_PIN, LOW);
            }
        }
    }

    // Shut down radio and enter Deep Sleep for 1 Hour
    LoRa.end();
    esp_sleep_enable_timer_wakeup(3600000000ULL);
    esp_deep_sleep_start();
}

void loop() {}

```

Node B: Indoor Receiver Gateway (Plugged into Raspberry Pi)

Upload this to the ESP32 that sits inside your home, connected via USB to your Home Assistant server.

```
#include <SPI.h>
#include <LoRa.h>

#define SS_PIN    7
#define RST_PIN   2
#define DIO0_PIN  3

void setup() {
    Serial.begin(115200);
    while (!Serial);

    LoRa.setPins(SS_PIN, RST_PIN, DIO0_PIN);
    if (!LoRa.begin(433E6)) {
        while (1);
    }

    LoRa.setSpreadingFactor(10);
    LoRa.setSignalBandwidth(125E3);
}

void loop() {
    // 1. Listen for Rooftop Data
    int packetSize = LoRa.parsePacket();
    if (packetSize) {
        String incomingData = "";
        while (LoRa.available()) {
            incomingData += (char)LoRa.read();
        }
        // Send data out the USB port to Home Assistant
        Serial.println(incomingData);
    }

    // 2. Listen for Home Assistant Commands
    if (Serial.available() > 0) {
        String haCommand = Serial.readStringUntil('\n');
        haCommand.trim();
        if (haCommand == "RUN_ROOF_IRRIGATION") {
            LoRa.beginPacket();
            LoRa.print("START_WATERING");
            LoRa.endPacket();
        }
    }
}
```

```
}
```

Phase 4: Home Assistant YAML Integration

Add this directly to your configuration.yaml on the Raspberry Pi. This intercepts the raw serial data over the USB port, filters it for the ROOF_NODE prefix, and creates a switch to trigger the watering cycle manually or via automation.

```
sensor:
  - platform: serial
    serial_port: /dev/ttyUSB0
    baudrate: 115200
    name: "Rooftop LoRa Raw Receiver"

template:
  - sensor:
      - name: "Rooftop Plants Moisture"
        unit_of_measurement: "%"
        state_class: measurement
        device_class: humidity
        state: >
          {% if 'ROOF_NODE:' in
states('sensor.rooftop_lora_raw_receiver') %}
          {{
states('sensor.rooftop_lora_raw_receiver').split('ROOF_NODE:')[1] |
int }}
          {% else %}
          {{ states('sensor.rooftop_plants_moisture') }}
          {% endif %}

switch:
  - platform: template
    switches:
      rooftop_irrigation_valve:
        friendly_name: "Rooftop Irrigation Valve"
        value_template: "off"
        turn_on:
          service: shell_command.send_lora_water_cmd
        turn_off:
          service: logger.log
          data:
            message: "Solenoid auto-shuts off via hardware timer"

shell_command:
  send_lora_water_cmd: "echo 'RUN_ROOF_IRRIGATION' > /dev/ttyUSB0"
```

Phase 5: Final Rooftop Physical Layout Rules

To ensure the system works flawlessly with the included spring antenna and survives the weather:

1. **Spring Antenna Placement:** The RA-02 module and its copper spring antenna must be positioned on the side of the junction box that faces your home. Keep it as far away from the LM2596 and MT3608 converters as possible to avoid electromagnetic interference. Do not stretch or deform the spring.
2. **No Metal Blockades:** Ensure your four 18650 batteries are not positioned directly between the spring antenna and your house building.
3. **Drip Loops & Glands:** Tighten the PG9 cable glands firmly and ensure the solar wire, sensor wire, and solenoid wire all feature a "U-shaped" drip loop outside the box so water cannot travel into the seals.
4. **Conformal Coating:** Paint the final soldered ESP32 board and the top of the moisture sensor with acrylic conformal coating to block humidity condensation.

This completes your master plan. You are officially ready to procure the hardware and build a professional-grade, 5-year LoRa irrigation node!